Work Breakdown Structure

for

AI Smart Assistant

Prepared by Team SENTINELS

Department of Computer Engineering,
University of Peradeniya

February 20, 2026

Contents

1 Introduction

This document provides a detailed Work Breakdown Structure (WBS) for the AI Smart Assistant System. The WBS organizes and defines the total scope of the project by decomposing it into manageable work packages.

2 Work Breakdown Structure Hierarchy

2.1 Level 1: AI Smart Assistant Project

The project is divided into seven major phases.

2.2 Level 2: Major Deliverables

1. **1.0 Project Management**
2. **2.0 Requirements Analysis**
3. **3.0 System Architecture & Design**
4. **4.0 Core Intelligence Engine**
5. **5.0 User Interface & Experience**
6. **6.0 Verification & Validation**
7. **7.0 Deployment & Final Delivery**

3 Detailed Decomposition (Level 3)

WBS ID	Work Package Description
1.0 Project Management	Management of timelines, resources, and stakeholder communication.
1.1 Planning	Creation of project charter, schedule, and resource allocation.
1.2 Risk Management	Identifying technical risks (OCR accuracy, LLM latency).
2.0 Requirements	Finalizing the SRS and user stories.
2.1 Stakeholder Interviews	Gathering specific needs from novice and power users.
2.2 Requirement Analysis	Defining REQ-SC, REQ-OCR, and REQ-AI specifications.
3.0 System Design	High-level and detailed design documents.
3.1 UX Design	Designing the non-intrusive transparent overlay.
3.2 Backend Architecture	Designing IPC between UI and AI Inference Server.
4.0 Core Engine	Development of the intelligence layer.
4.1 OCR Optimization	Implementing Tesseract/EasyOCR with GPU acceleration.
4.2 LLM Integration	Setting up Ollama/vLLM endpoints and prompt engineering.
5.0 UI Development	Implementation of the desktop frontend.
5.1 Overlay Rendering	Creating high-performance overlays using Win32/X11 APIs.
5.2 Interaction Logic	Mapping OCR regions to interactive UI elements.
6.0 Testing	Quality assurance and user acceptance.
6.1 Unit Testing	Testing individual OCR and AI modules.
6.2 Integration Testing	End-to-end testing from screen capture
7.0 Deployment	Final release and documentation.
7.1 Documentation	Finalizing User Manual and Troubleshooting Guide.
7.2 Installation Scripts	Creating installers for Windows.

Work Breakdown Structure

AI_Smart_Assistant_WBS

Friday 20 February 2026

WBS	Task	Stakeholders	Owner	Status	Due	Notes
1	**AI Smart Assistant Project**	**All Team Members, Supervisor**	**Team**	**In Progress**	**31/03/2026**	**Main project overview**
1.1	Project Management	Project Manager, Supervisor	Team	Completed	18/01/2026	Planning and risk management
1.1.1	Planning	Project Manager	Team	Completed	18/01/2026	Project charter, schedule, tool selection
1.1.2	Risk Management	Project Manager	Team	Completed	18/01/2026	Identify technical risks (OCR, LLM latency)
1.2	Requirements	All Team Members	Danuja	Completed	01/02/2026	Finalize SRS and user stories
1.2.1	Stakeholder Interviews	Novice & Power Users	Ashen	In Progress	25/01/2026	Gather needs from different user types
1.2.2	Requirement Analysis	All Team Members	Sooriya	Completed	01/02/2026	Define REQ-SC, REQ-OCR, REQ-AI
1.3	System Design	All Team Members	Janith	In Progress	08/02/2026	High-level and detailed design
1.3.1	UX Design	UX Lead	Janith	In Progress	08/02/2026	Design transparent overlay
1.3.2	Backend Architecture	Backend Lead	Ashen	Completed	08/02/2026	IPC between UI and AI server
1.4	Core Engine	AI Lead	Sooriya	In Progress	22/02/2026	Develop intelligence layer
1.4.1	OCR Optimization	AI Lead	Ashen	In Progress	15/02/2026	Implement Tesseract
1.4.2	LLM Integration	AI Lead	Sooriya	In Progress	22/02/2026	Set up Ollama/vLLM endpoints and prompts
1.5	UI Development	Frontend Lead	Janith	In Progress	15/02/2026	Implement desktop frontend
1.5.1	Overlay Rendering	Frontend Lead	Janith	In Progress	15/02/2026	High-performance overlay with Win32/X11 APIs
1.5.2	Interaction Logic	Frontend Lead	Team	In Progress	15/02/2026	Map OCR regions to interactive UI elements
1.6	Testing	QA Lead	Team	In Progress	01/03/2026	Quality assurance and user acceptance
1.6.1	Unit Testing	QA Lead	Team	In Progress	22/02/2026	Test individual OCR and AI modules
1.6.2	Integration Testing	QA Lead	Danuja	In Progress	01/03/2026	End-to-end testing from screen capture
1.7	Deployment	All Team Members	Danuja	In Progress	31/03/2026	Final release and documentation
1.7.1	Documentation	All Team Members	Danuja	In Progress	31/03/2026	Finalize user manual and troubleshooting guide
1.7.2	Installation Scripts	All Team Members	Team	In Progress	31/03/2026	Create installers for Windows